

Towards Automation and Reproducibility for Research Data Meshes

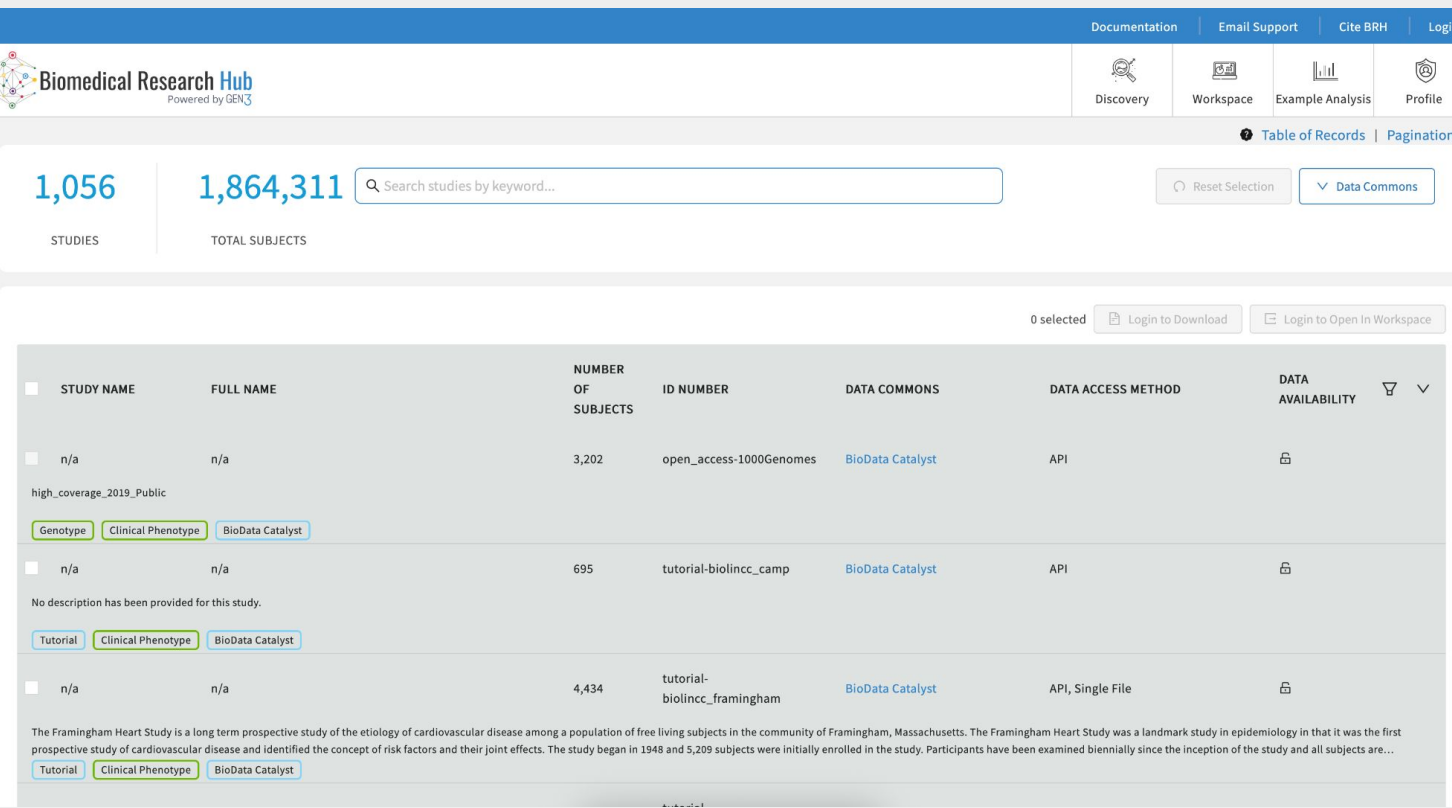
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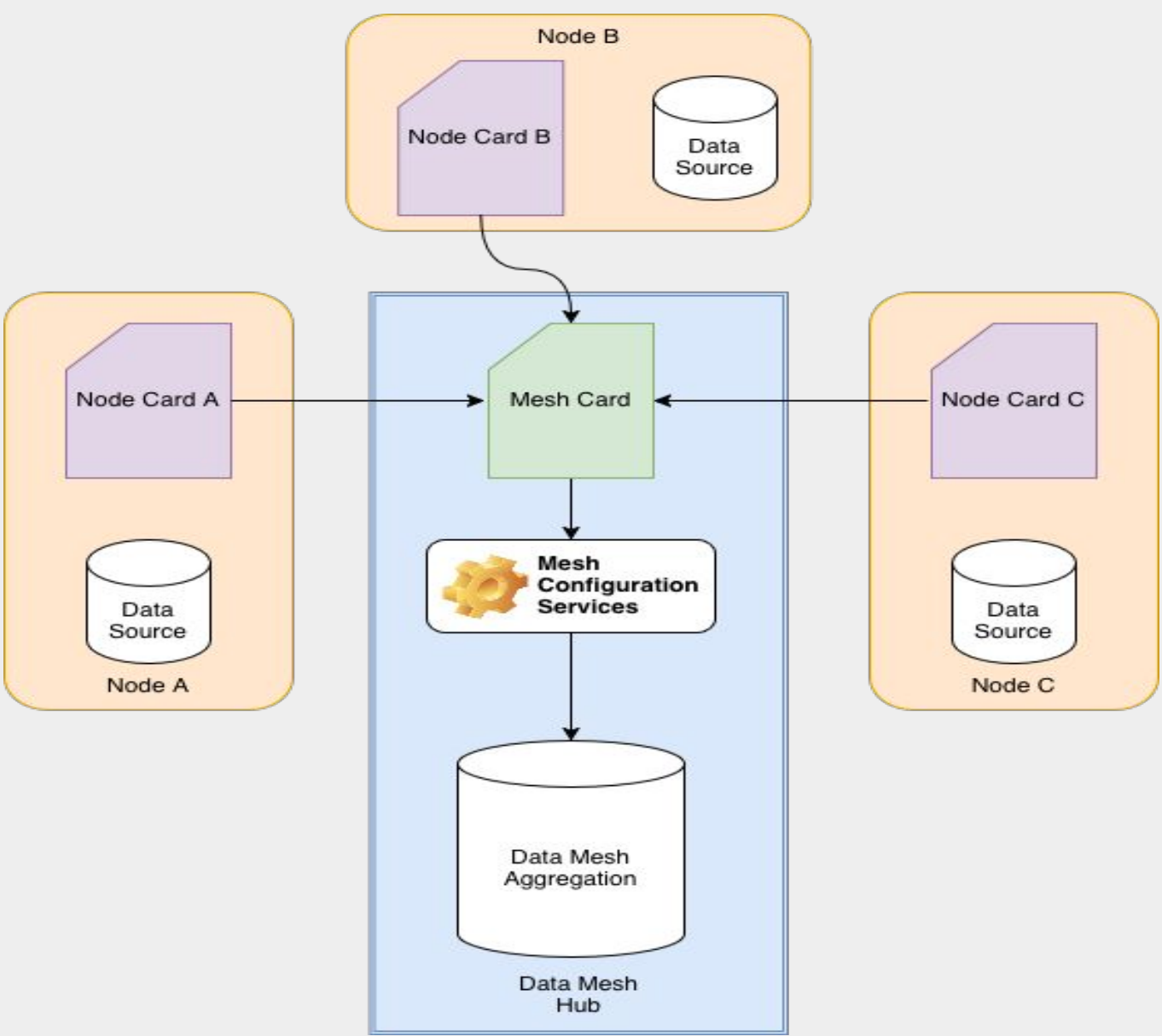
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Motivation



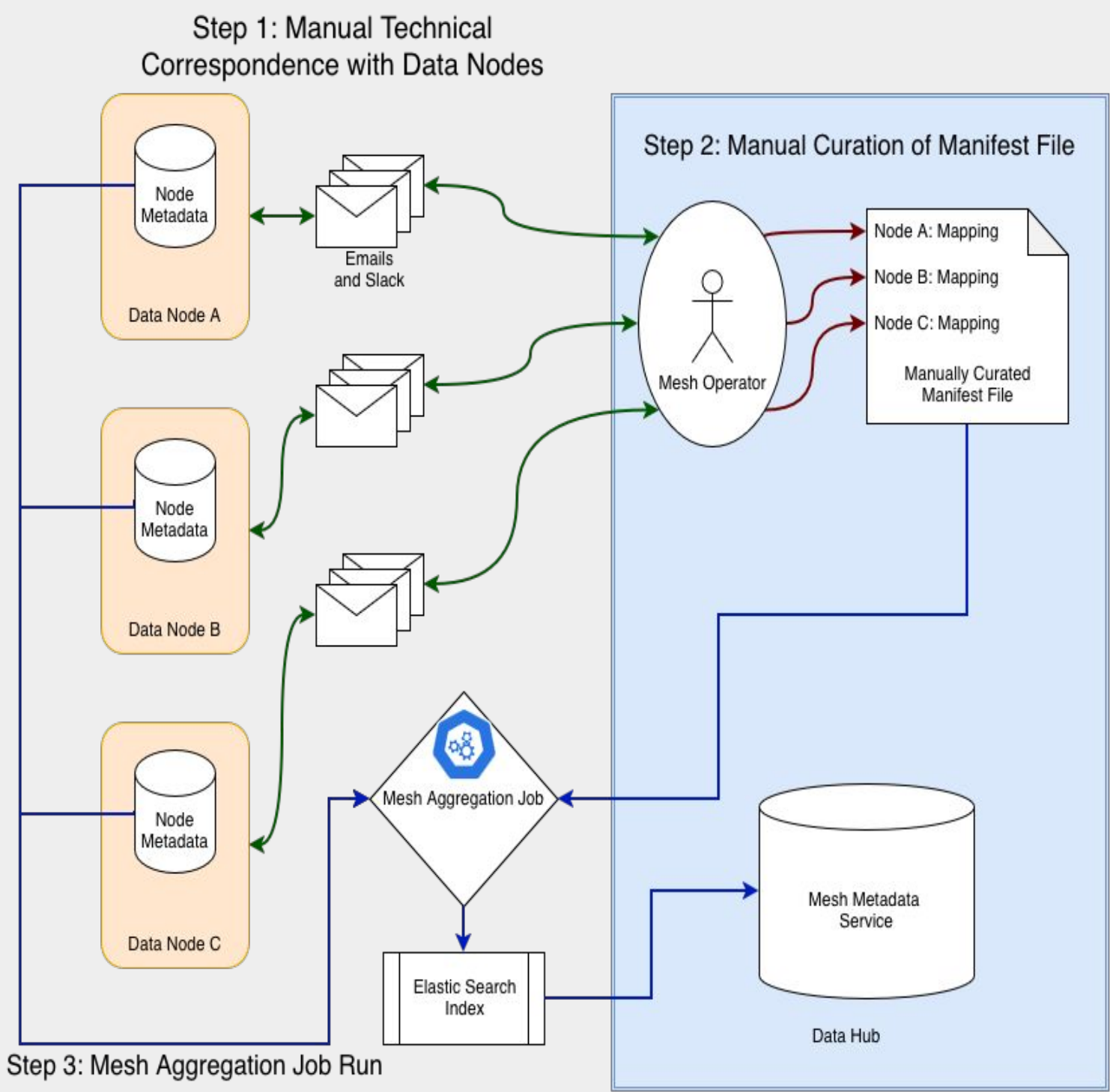
Research Data Meshes enable two or more data platforms (nodes) with different architectures operated by different organizations to interoperate. Examples of data meshes include 2 or more data commons that interoperate. It often took a month or more to add a new node to a data mesh since it has been a manual operation. We introduce node & mesh cards and supporting services that support the configuration, automation, and reproducibility of data meshes.

Setting up a Data Mesh

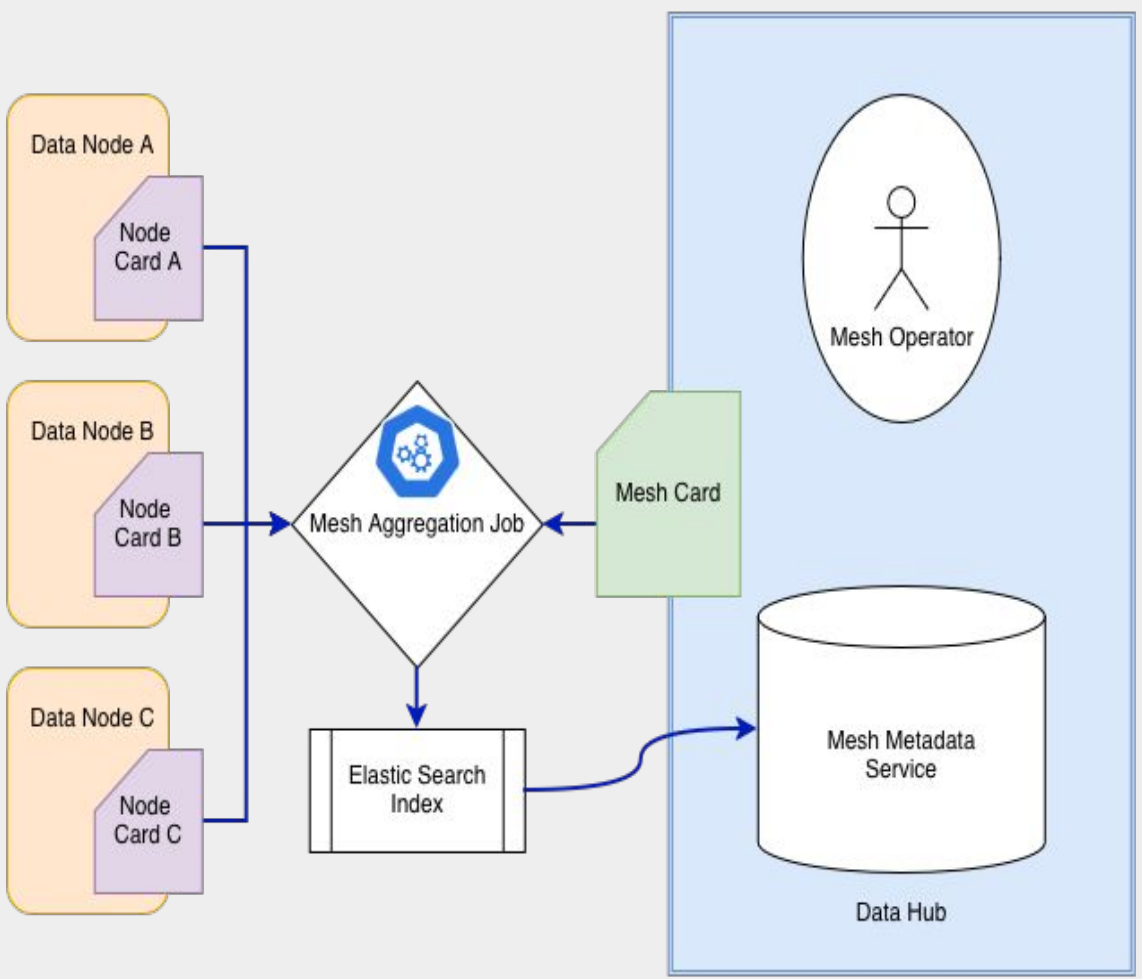


Using mesh and node cards we can easily automate setting up a data mesh.

Automation of Data Meshes



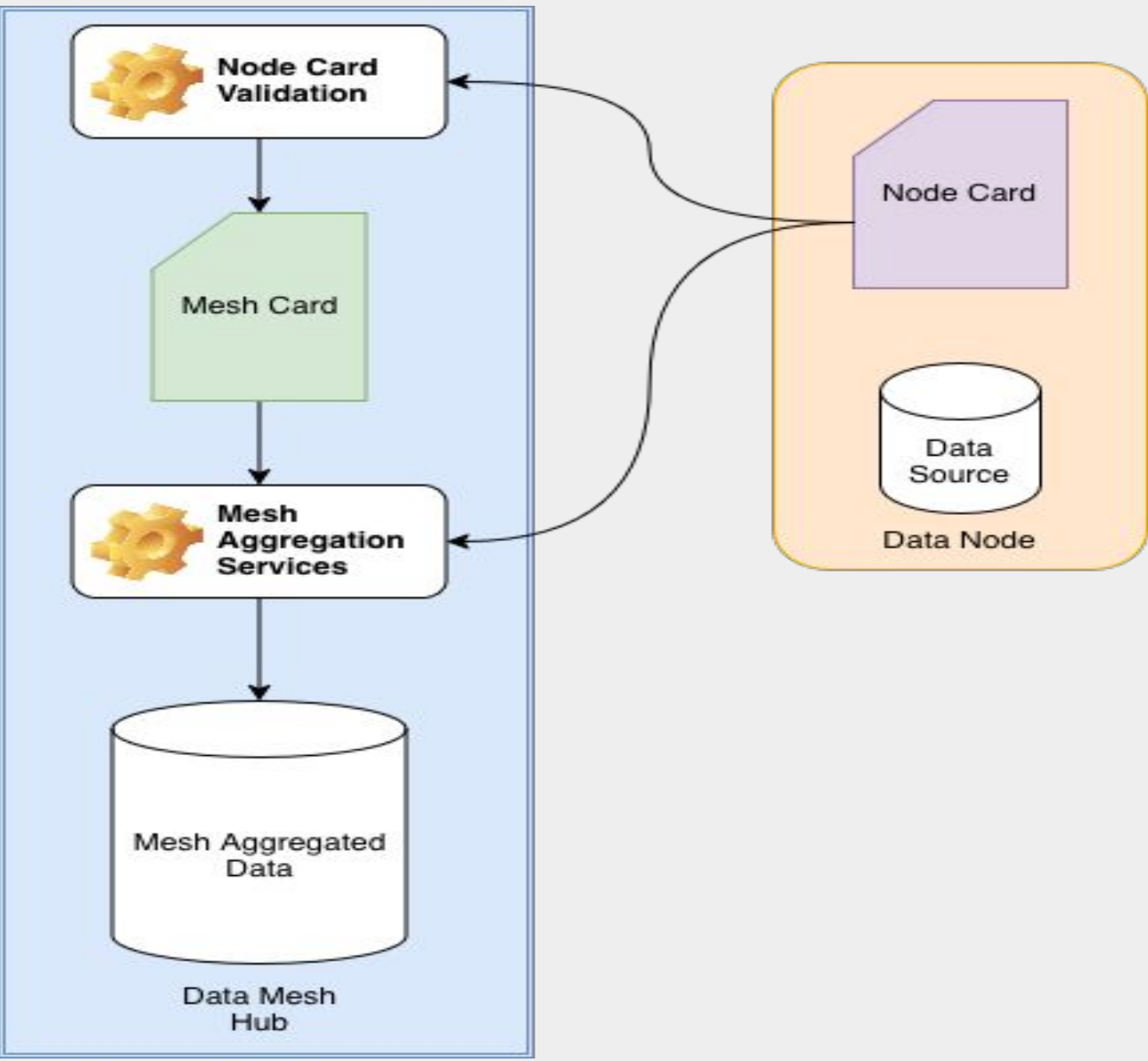
Without Mesh and Node Cards



With Mesh and Node Cards

Using mesh and node cards a mesh operator is able to automate orchestration of tasks such as aggregating node metadata. Previously, without mesh and node cards, manual correspondence and curation was required. With node and mesh cards validation of nodes is automated, and jobs have a central orchestration point

Node Joining a Mesh



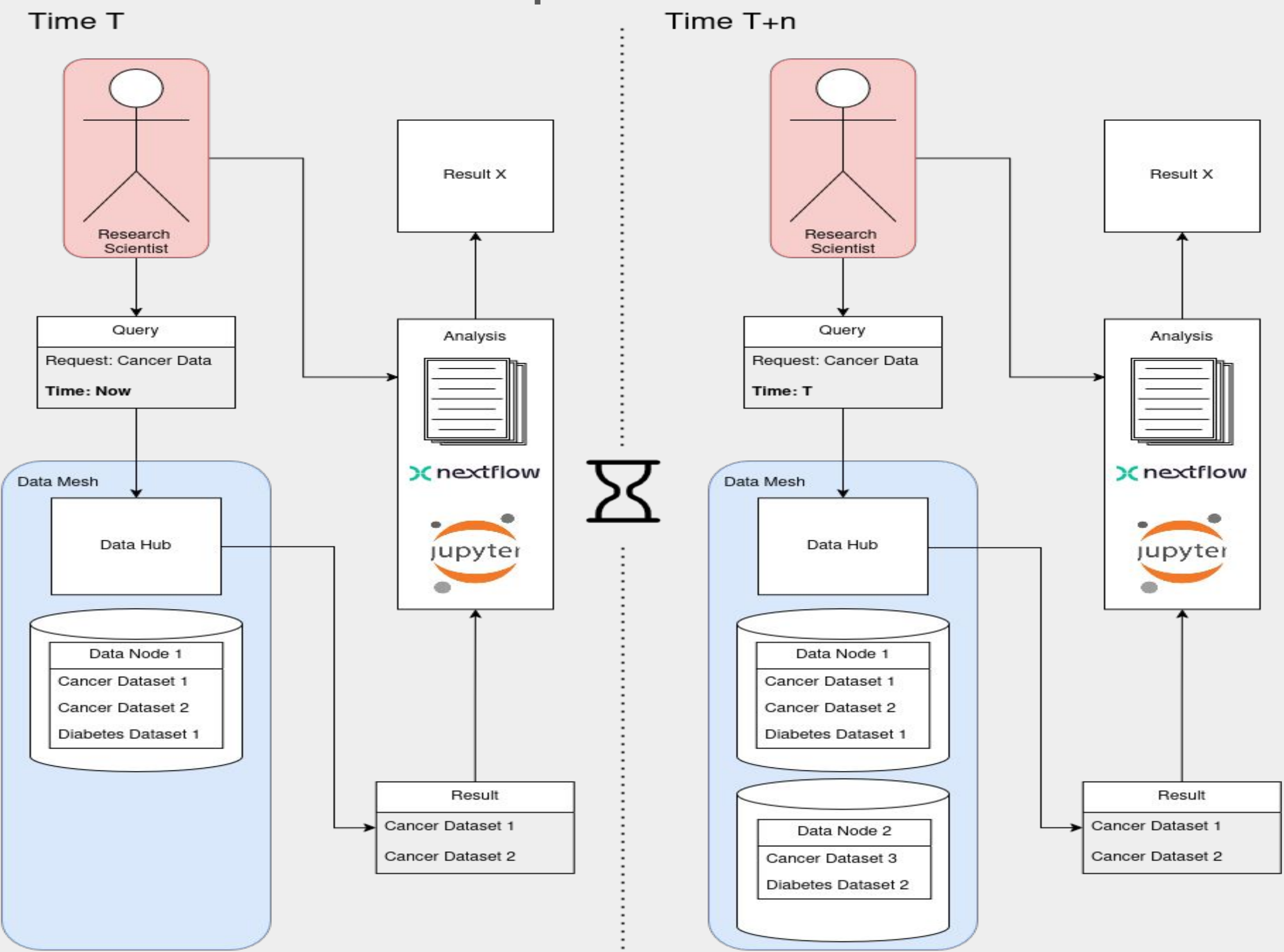
Nodes can easily join existing data meshes with a valid node card. Node operators create a node card that encapsulates the APIs required for interoperation in the mesh. That node card is then submitted to the mesh card API on the data mesh.

Acknowledgements

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Reproducible Data in a Mesh



Data in a typical data mesh is constantly being added and occasionally being dropped. Data meshes need services that can easily characterize the data available at a particular time used for a particular analysis in order to support reproducible research. Using a publish-subscribe methodology, we can efficiently define mesh state and support reproducible querying.